Global Edge Information Guidance With Polarity-Aware Attention for Satellite Object Detection

Huanxu Li, Keke Xu, and Xianglei Liu

***Abstract*—Object detection in remote sensing imagery remains challenging due to extreme-scale variations and deployment computational inefficiency. Existing Transformer-based detectors exhibit incomplete feature representations stemming from limited scale-awareness in handling multiscale objects, insufficient modeling of negative-polarity interactions, and uniform attention distributions. To address these limitations, this article proposes a novel framework integrating a global edge information extraction (GEIE) module and polarity-aware linear attention-based internal feature interaction (PLAIFI) module. The GEIE module consists of two components to integrate multiscale edge features: a polyresolution edge feature producer (PREFP) module and an edge fusion module. The PREFP module extracts edge information using a 3-D SobelConv branch applied to shallow features and generates a multiscale feature map to augment the hybrid encoder's multiscale representations with edge-aware priors. The edge fusion module supports flexible integration to enrich boundary-aware representations. To establish complementary feature enhancement, the PLAIFI module decomposes query–key pairs into positive-/negative-polarity components via learnable projection matrices, explicitly models same- and opposite-signed interactions through dual-polarity branches with polarity-sensitive coefficient matrices, and employs channelwise learnable power functions to sharpen attention distributions by reducing entropy—enhancing scale-awareness while maintaining linear complexity. Extensive experiments demonstrate state-of-the-art performance: 95.0% mean average precision (mAP) (+4.6%) on HRSC2016 (small-scale ships), 73.0% mAP (+5.9%) on DOTAv1.5 (multiclass objects), and 93.9% mAP on SIRSTv2 (tiny-scale infrared targets), achieved with 11.4% giga floating-point operations per second (GFLOPs)/11% Params increase over baseline real-time detection Transformer (RT-DETR). The principal innovation constitutes a synergistic framework where edge-aware feature recalibration and polarity-driven attention refinement mutually reinforce each other, establishing new state-of-the-art performance while maintaining deployment-friendly computational efficiency on satellite imagery. The source code will be available at: https://github.com/Keykeykeykeykey/Global_Edge_Guidance**

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I. Introduction

OBJECT detection involves identifying the category and location of various objects within an image, which is essential for applications such as environmental monitoring [1], [2], [3], [4], infrastructure assessment [5], [6], [7], [8], [9], and transportation management [10], [11], [12], [13], [14], [15]. This field has gained significant attention in remote sensing and computer vision [16], [17], [18], [19], [20], presenting considerable challenges. Developing robust object detectors for satellite imagery requires addressing the fundamental tradeoff between model accuracy and computational complexity, a challenge that becomes particularly pronounced in edge computing environments with limited resources.

Modern object detection frameworks are broadly categorized into two-stage or one-stage methods. Two-stage detectors, such as Mask R-CNN [21], leverage region proposal networks to first identify candidate regions and subsequently refine their classification and localization, achieving high accuracy for tasks like small bird detection [22]. One-stage detectors, exemplified by YOLOv5, unify localization and classification into a single step, enabling real-time performance in traffic surveillance [23], [24], [25] and pedestrian detection [26], [27], [28]. Recent advancements include hybrid architectures like YOLOv11 [29], [30], [31], [32], [33], [34], which integrate Transformer-based models like the Swin Transformer [35], which enhance the feature extraction through attention mechanisms. The introduction of Transformer architectures has revolutionized object detection paradigms. Detection Transformer (DETR) [36], [37], [38], [39] pioneered the application of Transformer-based models by reformulating object detection as a set prediction problem, eliminating the need for hand-crafted anchors or post-processing steps like nonmaximum suppression. However, DETR's computational complexity and slow convergence limited its practicality for remote sensing tasks requiring high-resolution image processing. To address these limitations, subsequent research introduced real-time DETR (RT-DETR), which optimizes the encoder–decoder structure by integrating hybrid-scale features and adaptive training strategies. RT-DETR [40] achieves significant improvements in both inference speed and detection accuracy for dense and small objects in remote sensing imagery, making it more suitable for large-scale applications. Other works, such as Swin Transformer [35], enabling efficient processing of high-resolution remote sensing data.

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These methods collectively demonstrate the adaptability of Transformer architectures to the unique challenges of remote sensing, including small object detection, orientation diversity, and complex backgrounds.

Despite these advancements, existing methods for object detection in satellite remote sensing imagery face two critical limitations. First, conventional multiscale feature integration mechanisms often struggle to balance detection performance across large and small objects, primarily due to insufficient exploitation of information across different backbone layers. While hybrid encoder and their variants attempt to fuse hierarchical features, they typically neglect the explicit transfer of global edge information between shallow and deep layers. This results in blurred boundary localization and degraded performance on objects with extreme scale variations—a particularly acute issue in high-resolution aerial imagery where targets coexist at vastly different scales. Second, while Transformer-based architectures leverage attention mechanisms (e.g., adaptive internal feature interaction (AIFI) module) for intrascale interaction, their single-layer Transformer encoders suffer from two fundamental limitations in satellite remote sensing scenarios as follows.

1) Information loss from polarity ignorance—existing linear attention approximations discard negative-value interactions in query–key pairs through nonnegative feature maps (e.g., ReLU), resulting in incomplete feature representations. This deficiency severely impairs foreground–background differentiation in complex scenes characterized by low interclass variance (e.g., ships versus wave textures).
2) Degraded spatial discriminability—the uniform attention distribution induced by high entropy in conventional kernel approximations diminishes the model's capacity to localize densely packed or partially occluded targets. These issues are further amplified by the unique characteristics of satellite imagery, including arbitrary object orientations (e.g., rotated vehicles in parking lots) and multiscale target distributions, ultimately leading to suboptimal detection accuracy and deployment computational inefficiency.

Edge features in object detection provide critical information about object boundaries, shapes, and contextual relationships, enhancing accuracy and robustness in identifying and localizing targets [41]. Recent studies have demonstrated the effectiveness of integrating edge features into deep learning frameworks for object detection. Jing et al. [42] proposed a feature aggregation network that employs a dual top–down pathway to enhance the spatial and semantic information, narrowing the semantic gap for small object detection through a feature-aware module. Zhu et al. [43] introduced GSTUnet, a gated-shaped Transformer-based architecture that combines Swin Transformer blocks with edge-aware gated convolutions to explicitly capture shape and edge information in infrared small target detection, achieving state-of-the-art performance. Xia and Yin [44] designed an improved deep network for liver detection that incorporates an edge-perception fusion module to enhance contour accuracy and a multiscale pyramid pooling layer for high-level feature extraction, significantly boosting detection performance. Lin et al. [45] further proposed a shape-biased learning framework to enhance robustness against varying target scales by integrating local edge priors with global semantic features. The integration of edge features into deep learning networks offers significant advantages, including enhanced sensitivity to target boundaries, improved robustness against low signal-to-clutter ratios, and reduced false alarms in complex backgrounds. By explicitly modeling edge and shape information, these methods mitigate the loss of fine-grained details during feature extraction, particularly critical for small and dim targets.

Prior research has sought to improve linear attention efficiency and expressiveness through various kernelized approximations. Katharopoulos et al. [46] introduced kernel-based feature maps to replace softmax, reducing quadratic complexity to linear with nonnegative ReLU activations. Qin et al. [47] proposed Cosformer, integrating ReLU with cosine reweighting to preserve locality inductive biases. Han et al. [48] utilized power operations on feature maps to approximate the exponential function while maintaining low entropy. Despite these efforts, existing methods suffer from information loss by neglecting negative query–key interactions and failing to restore softmax-like spikiness in attention distributions. In contrast, polarity-aware attention [49] addresses these limitations by explicitly modeling polarity-aware interactions (same-signed and opposite-signed components) via learnable power functions and sign-aware matrices, enabling sharper attention maps and comprehensive relationship modeling. When fused into deep learning architectures for object detection, polarity-aware attention improves fine-grained localization accuracy by capturing previously omitted negative-value interactions while retaining linear complexity, thereby enhancing robustness in distinguishing foreground–background features and bounding box regression.

We propose a global edge information extraction (GEIE) module to overcome the key flaw in satellite object detection: underexploited cross-layer edge semantics. Through low-coupling integration with feature pyramid layers, GEIE propagates edge information to high-level prediction networks by establishing a multiscale edge prior transmission mechanism. The module employs a Sobel operator-based poly-resolution edge feature producer (PREFP) on shallow features to capture critical boundary patterns while filtering background noise through preliminary convolutional processing. In hybrid encoder architectures, upsampled semantic features fused with edge details effectively recover boundary information lost during iterative downsampling, particularly crucial for maintaining the structural integrity of irregular-shaped targets in aerial imagery. Unlike prior edge-enhanced detectors [43], [45] that simply concatenate Sobel features, our GEIE module constructs a 3-D Sobel convolutional operator with scale-adaptive fusion coefficients, resolving extreme-scale variations through cross-layer edge semantics integration, while preserving deployment efficiency.

In addition, we propose a polarity-aware linear attention-based internal feature interaction (PLAIFI) module that enhances intrascale feature integration through three key mechanisms as follows.

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1) Polarity decomposition that preserves both same-signed and opposite-signed query–key interactions, crucial for discriminating subtle spectral differences in satellite imagery.
2) Dual-stream processing with channel-split value vectors and adaptive coefficient matrices, maintaining linear complexity while capturing spatial hierarchies.
3) Learnable power transformations that reduce entropy and restore attention sparsity through dynamic feature-dimension rescaling.

PLAIFI module supersedes RT-DETR's original AIFI module [40] by replacing its quadratic self-attention mechanism with a polarity-decomposed linear attention framework, that explicitly models positive/negative interactions through bipolar projections, enhancing intrascale feature integration while reducing computational complexity with entropy-reduced sharpening.

The contributions of this article are summarized as follows.

1) *GEIE Module Innovation:* We propose a GEIE framework that synergistically integrates multiscale edge features through 3-D Sobel operator fusion, which improves mean average precision (mAP) by 1.6% on small-scale targets (HRSC2016) and 4.6% on large-scale multiclass detection (DOTAv1.5) compared with baseline RT-DETR. The GEIE module addresses extreme-scale variations through cross-layer edge information integration, particularly improving low-contrast target localization compared to baseline methods (SIRSTv2).
2) *Polarity-Aware Attention Mechanism:* The PLAIFI module introduces a polarization-sensitive linear attention that adaptively weights feature importance based on spatial-semantic relationships, resolving the uniform attention distribution problem in conventional Transformers, which improves mAP by 3.1% on small-scale targets (HRSC2016) and 4.6% on large-scale multiclass detection (DOTAv1.5) compared with baseline RT-DETR with only 11.4% giga floating-point operations per second (GFLOPs) increase. The PLAIFI module addresses extreme-scale variations through polarity-enhanced intrascale feature integration, while maintaining parameter efficiency through entropy-reduced attention sharpening.
3) *Universal Detection Framework:* The framework establishes cross-domain superiority through synergistic integration of two core innovations: the GEIE module for multiscale boundary refinement and the PLAIFI for polarity-enhanced intrascale feature integration. The framework achieves state-of-the-art performance −73.0% mAP on DOTA-v1.5, 95.0% mAP on HRSC2016, and 93.9% mAP on SIRST v2 dataset. This unified architecture demonstrates robust generalization through adaptive edge-polarity co-learning, effectively addressing scale variance while maintaining parameter efficiency.

The remainder of this article is organized as follows. Section II introduces the proposed methodology. Section III discusses the experiment environment, experimental results, and comprehensive analysis. Section IV presents the future works. The conclusion is addressed in Section V.

II. Methodology

A. Overall of the Network

The overall proposed structure is shown in Fig. 1.

The proposed framework enhances the real-time DETR (RT-DETR) for object detection in satellite imagery through targeted architectural innovations. RT-DETR gained prominence for its real-time processing capability and superior accuracy in remote sensing object detection, offering an optimized foundation for enhancements. The backbone incorporates a GEIE module, which extracts shallow-layer edge features via a PREFP. These features are downsampled using max-pooling to preserve edge details and fused across scales through EdgeFusion (EFusion), enriching boundary-aware representations. In the Transformer encoder, the standard AIFI module is replaced with a PLAIFI. This mechanism captures nuanced query–key interactions using polarity-aware linear attention, while the cross-context feature fusion (CCFF) module aggregates multiscale contextual features to resolve spatial ambiguities. The decoder employs uncertainty-minimal query selection to refine object queries, reducing localization uncertainty, while the detection head generates bounding boxes and confidence scores. By integrating global edge guidance, polarity-driven attention, and uncertainty-aware decoding, the framework achieves robust detection of small objects in cluttered satellite scenes.

1) *Backbone With GEIE:* The GEIE module augments the CNN backbone. At shallow layers, the PREFP generates multiresolution edge maps from low-level features using parallel convolutions and max-pooling, retaining edge structures while suppressing background noise. These maps are hierarchically propagated across backbone stages and fused with backbone features via EFusion. This process employs cross-channel fusion, 3×3 convolution for local refinement and 1×1 convolution for dimension alignment, embedding edge-aware information into multiscale features to improve boundary localization. The parameter counts in the backbone are presented in Table I.
2) *Encoder With PLAIFI:* The encoder replaces the original AIFI with a polarity-aware linear attention mechanism that models same-signed and opposite-signed query–key interactions. This addresses information loss in traditional kernel-based linear attention, enhancing relational discrimination. The fusion module in CCFF (see Fig. 2) further aggregates multiscale features through adaptive channelwise attention and spatial pyramid operations, capturing long-range dependencies and cross-scale contexts essential for sparse small objects. A learnable power function sharpens attention maps by amplifying salient signals and suppressing noise, reducing entropy. The encoder outputs are processed by the decoder to generate final predictions.

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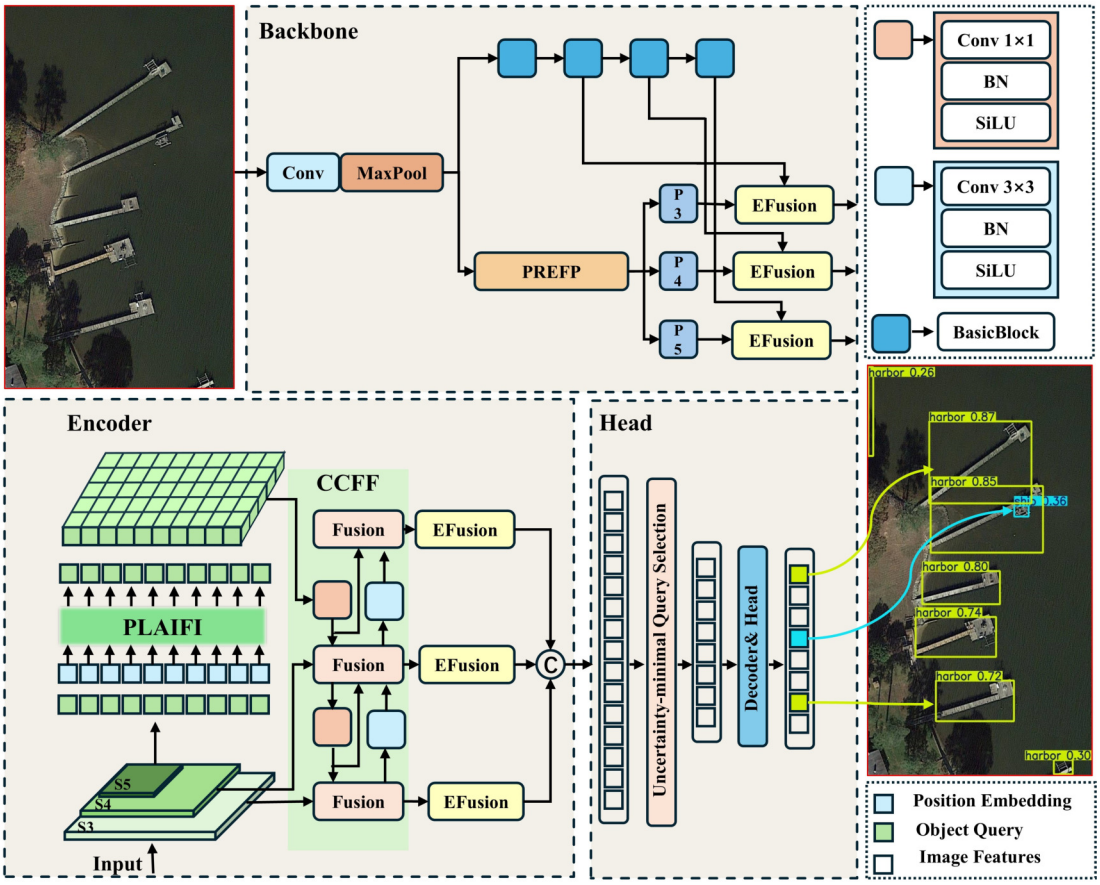


Fig. 1. Overall structure of the proposed network. Abbreviations: PREFP (Poly-Resolution Edge Feature Producer), PLAIFI (Polarity-aware Linear Attention based Internal Feature Interaction), and CCFF (Cross-Context Feature Fusion).

TABLE I
PARAMETER COUNTS IN BACKBONE

NO.	Module	Input	Output	Params	NO.	Module	Input	Output	Params
0	Conv	3	32	928	8	BasicBlock	64	64	152192
1	Conv	32	32	9280	9	BasicBlock	64	128	525568
2	Conv	32	64	18560	10	EFusion	[64,128]	128	57856
3	MaxPool	64	64	0	11	BasicBlock	128	256	2099712
4	PREFP	64	[64,128,256]	30720	12	EFusion	[128,256]	256	230400
5	P3	64	64	0	13	BasicBlock	256	512	8393728
6	P4	128	128	0	14	EFusion	[128,256]	512	919552
7	P5	256	256	0					

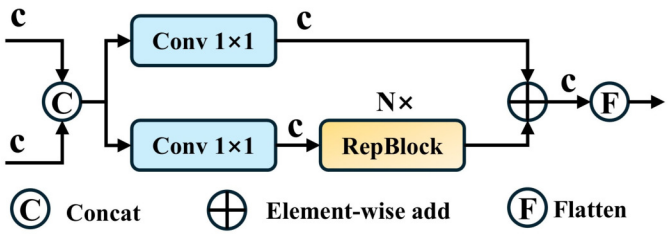


Fig. 2. Structure of the fusion module in CCFF. "C" denotes the cross-context feature component.

B. Algorithm Design of GEIE Module

The boundary localization accuracy of object detection models is frequently constrained by the limited sensitivity of high-level semantic features to edge details. To address this issue, this article proposes a GEIE module, which explicitly extracts edge features from shallow layers and dynamically fuses multiscale semantic information to significantly enhance the model's sensitivity to boundaries. The module comprises two core components: a PREFP mechanism and an adaptive fusion network, designed to hierarchically embed edge priors into backbone features for precise localization guidance.

1) Generation and Propagation of PREFP: The PREFP (see Fig. 3) module takes shallow features from the backbone network (e.g., the *P*2 layer) as inputs, which retain rich low-level edge responses while preliminarily suppressing background noise. Initially, the Sobel operator [50] is applied to compute gradient magnitudes of the feature maps, extracting initial edge contours.

The proposed SobelConv branch is constructed by embedding fixed Sobel operator kernels into a dual-path convolutional structure. The implementation consists of two key phases.

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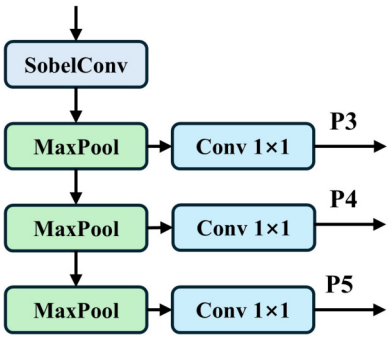


Fig. 3. Structure of PREFP.

a) Sobel kernel construction: We first define the horizontal (G_x) and vertical (G_y) Sobel operators as follows:

$$G_X = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{1}$$

$$G_Y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{2}$$

These 2-D kernels are then extended to 3-D convolutional weights through two steps (see Fig. 4).

1) *Channel Dimension Expansion:* Replicate each kernel along the channel dimension to match the input tensor's channel depth.
2) *Kernel Stacking:* Concatenate G_x and G_y along the output channel dimension to form a two-channel convolutional kernel.

The 3-D convolution in the SobelConv branch avoids computational redundancy and enhances efficiency by enabling parallel processing of multichannel feature maps directly from shallow layers, eliminating the need for raw-image 2-D Sobel operations.

b) Fixed-weight convolution: The constructed kernel W serves as nontrainable parameters for a 3-D convolutional layer. Specifically, during both forward and backward propagation

$$\frac{\partial L}{\partial W} = 0. \tag{3}$$

Parameter L is the loss function. This weight freezing guarantees preservation of gradient directionality while extracting edge features.

c) Feature fusion (see Fig. 5): The final edge-enhanced features are obtained through

$$(I * K)(x, y, z) = \sum_{m=-M}^{M} \sum_{n=-N}^{N} \sum_{p=-P}^{P} I(x+m, y+n, z+p) \cdot (m, n, p) \tag{4}$$

where I is the input feature map with dimensions m (depth), n (height), and p (width). K is the set of 3-D kernels. Here, M, N, and P denote the three dimensions, with m ranging from $-M$ to M, n from $-N$ to N, and p from $-P$ to P. x and y are the spatial positions in the output feature map.

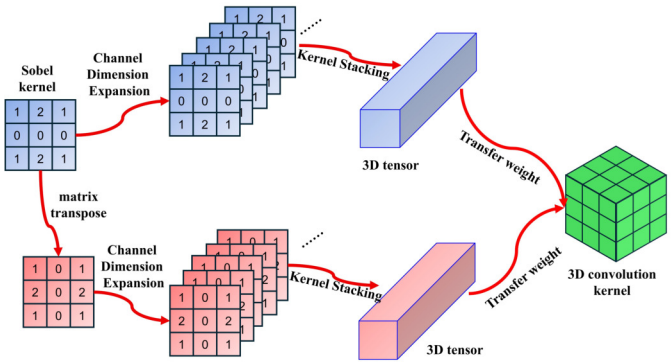


Fig. 4. 3-D convolutional kernel.

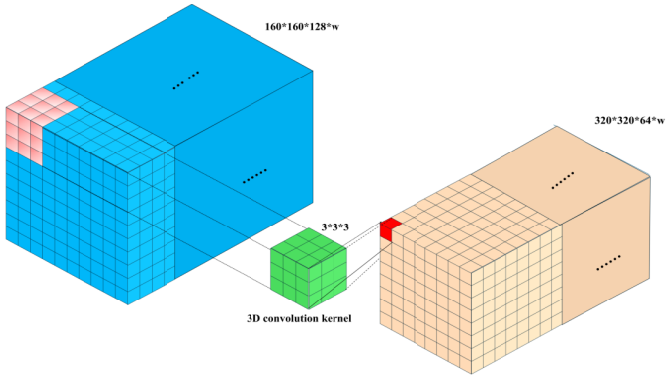


Fig. 5. 3-D group convolution.

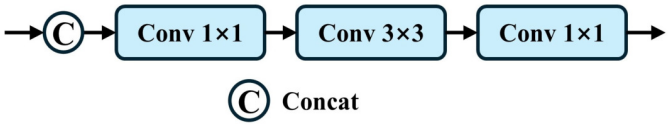


Fig. 6. Structure of EFusion Module.

Subsequently, multistage max-pooling layers generate downsampled edge features (8×, 16×, and 32×) to directly match the pyramid hierarchy ($P3$–$P5$) of the detection network. Compared to traditional edge extraction methods, this design leverages the noise-filtering capability of shallow convolutional layers, effectively mitigating interference from complex backgrounds in raw images.

2) Edge Feature Fusion Module: To ensure compatibility between edge features and backbone semantics, GEIE introduces an adaptive fusion architecture named EFusion (see Fig. 6). First, a 1×1 convolutional layer aligns the channel dimensions of concatenated edge and backbone features, dynamically balancing their weight distributions. A 3×3 depthwise separable convolution then reinforces local spatial correlations between edges and semantics, suppressing redundant information. This module supports flexible integration at the backbone output, hybrid encoder layers, or detection head inputs, enabling multilevel edge enhancement.

3) Polarity-Aware Linear Attention-Based Internal Feature Interaction: The single-layer Transformer encoder in AIFI struggles to capture multiscale dependencies in high-resolution satellite imagery due to limited hierarchical representation capacity, while its quadratic computational complexity and

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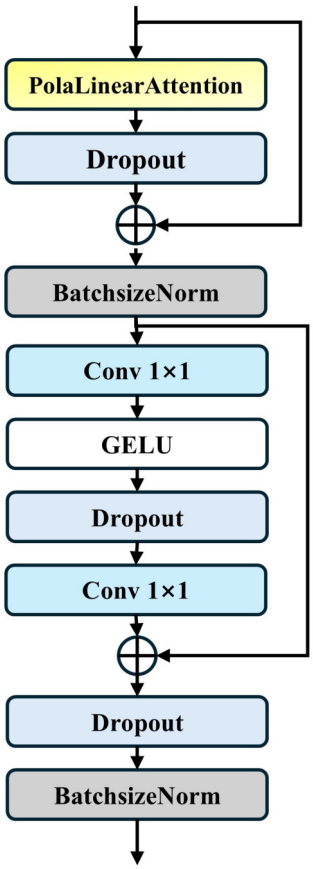


Fig. 7. Structure of PLAIFI.

nonnegative activation constraints inefficiently process high-resolution scenes. Hence, we proposed a module named PLAIFI to get a balance between expressiveness and efficiency. The structure is shown in Fig. 7.

The PLAIFI implements a vision-optimized Transformer architecture that processes feature maps through dual complementary pathways as follows.

1) *Polarity Attention Pathway:* Implements efficient spatial–channel interaction using polarity-aware linear attention mechanism, which operates on flattened features with localized window attention. This design reduces the quadratic complexity of standard self-attention while maintaining spatial coherence through constrained receptive field interactions.
2) *Convolutional Feed-Forward Network:* Replaces conventional linear projections with 1×1 convolutions, forming a bottleneck expansion structure that preserves spatial dimensions. This implementation maintains 2-D positional relationships while enabling channelwise nonlinear transformations through GELU activation and dropout regularization.

The polarity-aware linear attention [49] (as shown in Fig. 8) addresses these limitations through two synergistic innovations as follows.

1) Polarity-Driven Decomposition: Queries and keys are decomposed into positive and negative components

$$q = q^+ - q^- \tag{5}$$

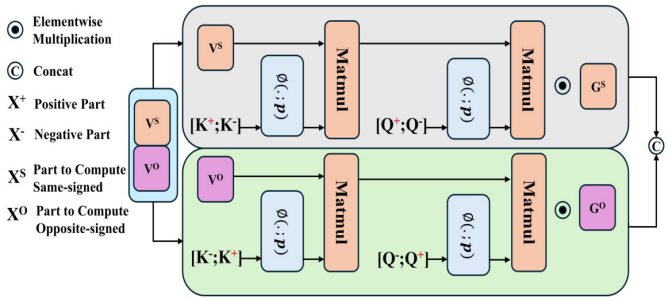


Fig. 8. Structure of polarity-aware linear attention.

$$k = k^+ - k^- \tag{6}$$

where $q^+ = \text{RELU}(q), q^- = \text{RELU}(-q), k^+ = \text{RELU}(k)$, and $k^- = \text{RELU}(-k)$.

Existing traditional linear attention methods neglect negative components, limiting relational modeling. To comprehensively cover all polar interactions, polarity-aware linear attention explicitly computes the approximate sum of four types of interactions

$$\begin{aligned}\text{Softmax} \approx \Phi\left(q^+\right)\Phi\left(k^+\right)^T + \Phi\left(q^-\right)\Phi\left(k^-\right)^T \\ - \Phi\left(q^+\right)\Phi\left(k^-\right)^T - \Phi\left(q^-\right)\Phi\left(k^+\right)^T\end{aligned} \tag{7}$$

where Φ is the feature mapping function.

To avoid the instability of subtraction operations, the weights of each component are dynamically adjusted through a learnable polarity-aware matrixGs (processing interactions with the same sign) and Go (processing interactions with opposite signs).

A polarity-driven decomposition that explicitly models both same-signed and opposite-signed query–key interactions through dual-stream processing. By separating positive/negative components and recombining them via learnable sign-aware matrices, it comprehensively reconstructs the original dot-product relationships while maintaining linear complexity.

2) Entropy Elevation: To restore a "sharp" attention distribution similar to Softmax, polarity-aware linear attention designs a learnable channel constrained by strictly increasing ($g' > 0$) and convex ($g'' > 0$) properties to reduce the attention distribution entropy

$$g(x; p) = x^{p_i} \tag{8}$$

where $p_i = 1 + \alpha \cdot \text{sigmoid}(\omega_i)$, α is a hyperparameter, and ω_i is learnable, enabling task $\phi(x^+)$ and $\phi(x^-)$, effectively distinguishing strong and weak attention signals to generate more focused attention weights.

Entropy suppression via dimensionwise power functions with learnable exponents, which act as differentiable sharpening operators. These functions, constrained by positive first/second derivatives, systematically enhance attention spikiness by amplifying strong responses and suppressing weak ones, effectively restoring the low-entropy distributions characteristic of softmax attention without exponential operations.

This dual mechanism resolves AIFI's critical shortcomings: The polarity-aware design mitigates information loss from truncated negative interactions, enabling precise feature correlation modeling, while the entropy control recovers

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the "focus-and-exclude" behavior essential for task-specific saliency. Unlike localized or sparse approximations, PLAIFI preserves full-sequence modeling capacity while operating at linear complexity, making it ideal for long-range dependencies in high-resolution vision tasks. Its architectural agnosticism further allows seamless integration with existing attention-based frameworks, offering a principled balance between expressiveness and efficiency.

III. Results

A. Datasets

The DOTA dataset [51] is a comprehensive resource for aerial object detection. The DOTA-v1.5 version comprises 2806 images, with dimensions varying from 800 × 800 to 4000 × 4000 pixels. These images are sourced from diverse sensors and platforms, including Google Earth, Gaofen-2, and JL-1, and encompass multiple resolutions. The dataset categorizes objects into 15 main distinct classes: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), Ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Consistent with standard practice, we utilize both the training and validation sets for model training, while the testing set is reserved for evaluation. The division of images in the dataset includes 1411 for training, 458 for validation, and 937 for testing.

In contrast, the HRSC2016 dataset [52] is specifically designed for ship detection in aerial imagery. This dataset contains 1061 images, which include a total of 2976 ship instances. Among these, 70 images are dedicated to sea scenes, yielding 90 samples, while the remaining 991 images feature both sea and land, contributing 2886 samples. The sizes of images in HRSC2016 range from 300 × 300 to 1500 × 900 pixels, with most exceeding 1000 × 600 pixels. Similar to DOTA, HRSC2016 is divided into training, validation, and testing sets, comprising 436, 181, and 444 images, respectively.

Designed for single-frame infrared small target detection, the SIRST dataset (v2) [53] addresses challenges unique to dim, sparse targets in complex backgrounds. This dataset comprises 514 images containing a total of 480 target instances, with images meticulously curated from hundreds of infrared sequences to ensure no temporal overlap between subsets. The dataset is partitioned into training (70%, 360 images), validation (20%, 103 images), and testing (10%, 51 images) sets. All images are standardized to 300 × 300 pixels, with targets occupying as few as 0.02% of the image area (e.g., 3 × 3 pixels in a 300 × 300 image). Notably, 90% of images contain only one target, while approximately 55% of targets exhibit extremely low contrast, with 65% being dimmer than their background.

Together, these datasets provide a robust foundation for advancing object detection methodologies in aerial imagery, each catering to different aspects of the domain while ensuring a diverse set of scenarios and object types for comprehensive analysis.

TABLE II
Comparison Results on the HRSC2016 Dataset

Method	*mAP* (%)
R2CNN	73.07
RoI-Transformer	86.20
Oriented RCNN	90.50
RA2DCNet	89.75
DAL	89.77
ReDet	90.46
LSKNet-S	90.65
PKINet-S	90.70
YOLOv10n	94.30
CRAS-YOLO	90.50
YOLOv11s	91.60
RTDETR	91.00
Ours	95.00

B. Implementation Details

The initial learning rate is set at 0.01 and each model is trained for 500 epochs with batch size = 2. The input image is resize to 640 × 640, and the speed of all models is measured on a GeForce RTX 4070Ti Super GPU 16G and i7-12600kf CPU, using Pytorch 2.4.1, cuda 12.4 and cudnn 7.5.0.

C. Evaluating Metric

To assess the effectiveness of our proposed methods, we utilize several key performance indicators: precision, recall, mAP, and floating-point operations (FLOPs). Precision measures the proportion of correctly identified positive samples among all detected positives. In contrast, recall evaluates the proportion of actual positive samples that have been successfully identified. These metrics are defined as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{9}$$

In the above equation, true positives (TPs) refers to the number of positive samples correctly identified and false positives (FPs) indicates the number of negative samples incorrectly classified as positive.

We also compute the mAP to provide a comprehensive evaluation of our model's performance. The calculation of mAP is defined as follows:

$$\text{AP} = \int_0^1 P(R)\,\text{dR} \tag{10}$$

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i. \tag{11}$$

D. Comparison With SOTA

1) *HRSC2016:* We used 617 images in HRSC2016 dataset for training and validation. The results of R2CNN [54], RoI-Transformer [55], Oriented RCNN [56], RA2DCNet [57], DAL [58], ReDet [59], LSKNet-S [60], PKINet-S [61], CRAS-YOLO [62], YOLOv10n [63], YOLOv11s [64], and RT-DETR [40] are compared. Table II shows that comparison by mAP.As shown in Table II, our method achieves state-of-the-art performance with 95.00% mAP, surpassing all compared approaches. The baselines include: R2CNN (73.07%

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TABLE III
RESULTS OF EACH OBJECT CLASS ON THE DOTA-V1.5 DATASET

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	*mAP* (%)	Params (M)	GFLOPs	Inference speed (ms)
RetinaNet-O (2017)	71.4	77.6	42.1	64.7	44.5	56.8	73.3	90.8	76.0	60.0	47.0	69.2	59.7	64.5	48.1	59.2	35.7	299.5	-
RF. R-CNN (2016)	72.2	76.4	47.6	69.9	52.0	70.5	80.3	90.9	79.2	68.6	59.6	72.3	66.4	66.1	55.3	64.6	41.2	292.3	-
Mask R-CNN (2017)	76.8	73.5	49.9	57.8	51.3	71.3	79.8	90.5	74.2	66.1	46.2	70.6	63.1	64.5	57.8	62.7	40.1	207.2	-
HTC (2019)	77.8	73.7	51.4	64.0	51.5	73.3	80.3	90.5	75.2	67.3	48.5	70.6	64.8	64.5	55.9	63.4	80.2	441.3	-
RoI-Trans (2019)	72.3	82.0	54.5	70.0	52.5	76.3	81.0	90.9	84.2	69.1	62.9	72.3	68.7	65.9	57.1	66.7	55.1	200.2	-
ReDet (2021)	79.2	82.8	51.9	71.4	52.4	75.7	80.9	90.8	75.8	68.6	49.3	72.0	73.4	70.6	63.3	66.9	31.6	-	-
FRED (2024)	79.6	81.4	52.6	72.6	58.1	74.8	86.1	90.8	82.1	74.8	53.4	72.9	69.5	59.9	54.8	68.3	-	-	-
YOLO v11s (2024)	75.5	75.2	36.4	63.1	64.3	71.0	76.3	88.3	46.7	76.8	49.6	66.7	58.1	63.3	62.0	66.9	25.9	6.3	4.2
YOLO v12 (2025)	71.2	74.2	38.6	50.1	60.0	66.8	75.2	85.3	49.2	75.8	42.1	42.5	56.1	58.5	78.1	64.0	25.1	5.8	5.8
RT-DETR (2024)	86.8	80.8	48.4	50.4	64.6	79.2	80.6	91.7	72.2	74.6	44.9	71.6	69.8	74.8	82.8	67.1	19.9	57.0	9.6
Ours	86.6	70.2	43.9	67.6	61.9	77.3	79.7	94.8	75.7	66.0	60.4	69.1	71.7	76.1	66.1	73.0	22.0	63.5	7.4

mAP), RoI-Transformer (86.20%), Oriented RCNN (90.50%), RA2DCNet (89.75%), DAL (89.77%), ReDet (90.46%), LSKNet-S (90.65%), PKINet-S (90.70%), CRAS-YOLO (90.50%), and Transformer-based RT-DETR (91.00%). Notably, our approach outperforms YOLO-family variants including YOLOv10n (94.30%) and YOLOv11s (91.60%) by significant margins. The results validate the effectiveness of our architectural innovations for aerial object detection tasks.

2) *DOTAv1.5:* To verify the effectiveness of the algorithms in this article, we compare our method with ten state-of-the-art algorithms (RetinaNet [65], RF. R-CNN [66], Mask R-CNN [21], HTC [67], RoI-Transformer [55], ReDet [59], FRED [68], YOLO v11 [64], YOLO v12 [69], and RT-DETR [40]). The comparison results are shown in Table III.

The experimental results on the DOTA-v1.5 dataset demonstrate the superior performance of our proposed method, achieving a state-of-the-art mAP of 73.0%, surpassing all compared approaches, including RT-DETR (67.1%) and FRED (68.3%). Notably, our method exhibits remarkable improvements in boundary-sensitive categories: it attains the highest AP in (TC, 94.8%), (SP, 76.1%), and (SBF, 60.4%), outperforming previous best results by 3.1%, 1.3%, and 7.0%, respectively. The proposed GEIE module significantly enhances boundary localization accuracy, particularly evident in geometrically complex classes like SP (+1.3% versus RT-DETR) and TC (+3.1% versus RT-DETR). Furthermore, the polarity-aware linear attention mechanism contributes to robust feature representation across scale variations, as demonstrated by superior performance in small-object categories such as (HC, 66.1%), while maintaining computational efficiency.

Comparative analysis reveals our method's balanced multiscale detection capability: it achieves competitive results in large-area categories like plane (PL, 86.6%) and ship (SH, 79.7%) while substantially improving small-vehicle (SV, 61.9%) detection over rotation-based detectors like RoI-Trans (52.5%). The dynamic fusion strategy in GEIE enables effective cross-scale feature integration, particularly beneficial for heterogeneous objects like harbor (HA, 71.7%) with mixed land-sea boundaries. The value-preserving polarity-aware linear attention further reduces localization ambiguity in cluttered scenes, as evidenced by the 5.9% mAP gain over RT-DETR despite using a similar Transformer architecture.

Furthermore, our method achieves state-of-the-art mAP on DOTA-v1.5, while maintaining deployment-friendly efficiency: with only 22.0M parameters and 63.5 GFLOPs, it reduces inference latency to 7.4 ms—outperforming RT-DETR (19.9M/57.0GFLOPs/9.6 ms) in speed despite slightly higher computations, and significantly exceeding rotation-based models like RoI-Trans (55.1M/200.2GFLOPs) in efficiency. This demonstrates successful resolution of the accuracy-efficiency tradeoff emphasized: The GEIE module's edge-aware feature recalibration and PLAIFI's linear-complexity polarity modeling synergistically address scale variation and boundary ambiguity challenges without compromising real-time applicability, validating the framework's practicality for satellite imagery deployment as concluded.

TABLE IV
RESULTS OF EACH 3-D EDGE DETECTION OPERATOR ON HRSC2016 DATASET

Operator	*mAP* (%)
Sobel	92.6
Roberts	92.2
Prewitt	90.8

E. Ablation Study

1) *Ablation Study on PREFP Module:* To ascertain the optimal edge detection operator selection for the PREFP module design, we systematically evaluated three classical gradient-based operators (Sobel [50], Roberts [70], and Prewitt [70]) through 3-D convolutional kernel construction on HRSC2016 dataset. Each 3-D kernel synthesis rigorously followed two critical technical procedures: Channel dimension expansion to establish

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TABLE V
ABLATION RESULTS FOR PLAIFI BLOCK ON HRSC2016 DATASET

Operator	*mAP* (%)
AIFI (Original)	91.0
Polarity-aware Attention	94.1
TSSA	93.3

cross-channel correlation awareness, and kernel stacking for multioperator spatial response fusion, implemented via depthwise concatenation across the tensor's third dimension. The results are shown in Table IV. The systematic evaluation of 3-D edge detection operators (Sobel, Roberts, and Prewitt) on the HRSC2016 dataset, implemented via channel dimension expansion and kernel stacking fusion, demonstrated Sobel's superior performance with a 92.6% mAP, outperforming Roberts (92.2%) and Prewitt (90.8%). This advantage stems from Sobel's 3 $\times$ 3 kernel design, which integrates Gaussian smoothing for noise suppression while maintaining balanced directional sensitivity—critical for detecting continuous ship contours in complex maritime scenes. Roberts' 2 $\times$ 2 kernel, though efficient, exhibited fragility to noise and incomplete edge localization for large-scale targets, while Prewitt's lightweight design suffered from gradient magnitude dilution during fusion. Sobel's rotational symmetry and diagonal gradient templates aligned optimally with HRSC2016's dominant ship geometry, ensuring robust edge coherence under occlusion and viewpoint variations. The results validate Sobel as the optimal baseline for 3-D edge-aware detection tasks requiring precision-noise tradeoffs. Hence, we select the Sobel operator to design the PREFP module.

2) *Design of PLAIFI Block:* In the PLAIFI module design, we incorporated the latest attention algorithms (TSSA [71]) to replace the attention component of our proposed architecture, with experimental validation conducted on the HRSC2016 dataset. The results are shown in Table V. As shown in Table V, the results, measured in mAP, demonstrate significant performance variations: the baseline AIFI (original) achieved an mAP of 0.910, while polarity-aware attention yielded the highest improvement at 0.941, and TSSA attained 0.933. This indicates that incorporating polarity-aware attention substantially enhances feature interaction efficacy, likely due to its robust multiscale dependency modeling, whereas TSSA offers a moderate gain over the baseline but falls short of polarity-aware attention's superiority, suggesting that the specific architectural optimizations in polarity-aware attention are more effective for long-range context capture in this aerial target detection task.
3) *Ablation Study on DOTA v1.5:* The ablation studies in Table VI demonstrate the individual and combined effectiveness of proposed components on DOTA-v1.5 dataset. The baseline RT-DETR achieves 67.1% mAP without any enhancements. When separately incorporating GEIE or PLAIFI, the model gains identical improvements of +4.6% mAP (71.7%), suggesting both components independently address critical challenges in aerial object detection. Notably, their synergistic integration yields superior performance with 73.0% mAP (+5.9%), indicating complementary benefits: multiscale edge enhances boundary-aware feature representation across scales while PLAIFI adaptively models geometric variations through polarity-aware linear attention mechanisms. This cumulative improvement underscores the importance of jointly addressing multiscale feature extraction and orientation-sensitive object characteristics in remote sensing imagery.

TABLE VI
ABLATION RESULTS ON DOTA V1.5 DATASET

GEIE	PLAIFI	*mAP* (%)
		67.1
√		71.7 (+4.6)
	√	71.7 (+4.6)
√	√	73.0 (+5.9)

TABLE VII
ABLATION RESULTS ON HRSC2016 DATASET

GEIE	PLAIFI	*mAP* (%)
		91.0
√		92.6 (+1.6)
	√	94.1 (+3.1)
√	√	95.0 (+4.0)

4) *Ablation Study on HRSC 2016:* The ablation results in Table VII reveal distinct contributions of proposed modules on HRSC2016 dataset. The baseline model achieves 91.0% mAP without enhancements. Incorporating the GEIE alone yields a moderate improvement of +1.6% mAP (92.6%), while the PLAIFI demonstrates more substantial gains of +3.1% mAP (94.1%), highlighting its critical role in handling orientation variations prevalent in maritime targets. The combined implementation achieves state-of-the-art performance at 95.0% mAP (+4.0%), where multiscale edge captures hierarchical boundary semantics to mitigate scale variance, and PLAIFI dynamically models directional features through polarization-sensitive attention. This performance hierarchy suggests that geometric-aware modeling dominates in oriented object detection, while edge-scale refinement provides auxiliary regularization. The results emphasize the necessity of dual-path feature learning for precise localization of rotation-sensitive instances in aerial/satellite imagery.

F. GFLOPs and Parameters

The computational complexity (in GFLOPs) and Parameters compared with each version RT-DETR, where "*r*" denotes ResNet-based backbone depth and "*x*" signifies an extreme-scale architecture, are shown in Table VIII.

Based on the data presented in Table VIII, our method incurs a moderate computational increase of 11.4% in GFLOPs (63.5 versus 57.0) and 10.6% in parameters (21.99M versus 19.89M) relative to the baseline RT-DETR-r18, while

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TABLE VIII
GFLOPS AND PARAMETERS COMPARISON WITH BASELINE

Method	GFLOPs	Parameters
RT-DETR-r18	57.0	19.89M
RT-DETR-r34	90.6	31.23M
RT-DETR-r50	129.9	42.12M
RT-DETR-l	108.3	32.97M
RT-DETR-x	232.7	67.47M
Ours	63.5	21.99M

TABLE IX
RESULTS ON THE SIRST V2 DATASET

Method	*mAP* (%)
U-Net [72]	80.30
ACM-FPN [53]	81.76
ACM-Unet [53]	80.78
ALCNet [73]	84.42
AGPC [74]	85.63
IAANet [75]	70.27
UIU-net [76]	82.41
MSAFFNet [77]	85.41
Ours	93.90

maintaining substantially lower complexity than larger variants. This lightweight complexity increase achieves significant accuracy gains: +5.9% mAP on DOTA-v1.5 (see Table VI) and +4.0% mAP on HRSC2016 (see Table VII), demonstrating robust generalization across datasets with diverse object scales and orientations. The results validate that our proposed modules (GEIE and PLAIFI) strike an optimal efficiency-effectiveness balance, delivering substantial performance improvements while maintaining practical deployability in resource-constrained aerial detection scenarios.

G. Detecting Infrared Tiny Targets

To evaluate the efficacy of the proposed method for detecting infrared tiny targets, we conducted a comparative analysis with eight state-of-the-art algorithms, as summarized in Table IX.

As shown in the Table IX, our method achieves state-of-the-art performance (93.90% mAP) on the SIRST v2 dataset, surpassing existing approaches by significant margins (e.g., +8.49pp over MSAFFNet). This superiority stems from our framework's tailored design for small targets: the GEIE module counters extreme target sparsity and significant target–background contrast challenges through multiresolution edge map fusion from shallow features, injecting explicit boundary priors to resolve detection ambiguities in cluttered environments. Concurrently, the PLAIFI module mitigates uniform attention distribution in conventional linear attention via polarity-driven interaction and entropy-reduced sharpening, enhancing discriminative power for subpixel targets (e.g., 3 $\times$ 3 pixels). The integrated edge-aware feature recalibration and attention refinement synergistically improve localization precision for single-target scenarios while suppressing false alarms. Our method also demonstrates consistent efficacy across datasets, achieving competitive results on DOTA-v1.5 as previously validated.

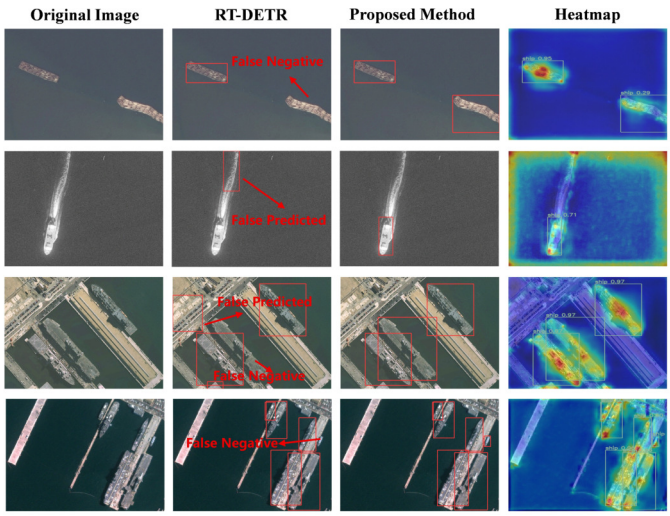


Fig. 9. Visualizations of detection results on HRSC2016.

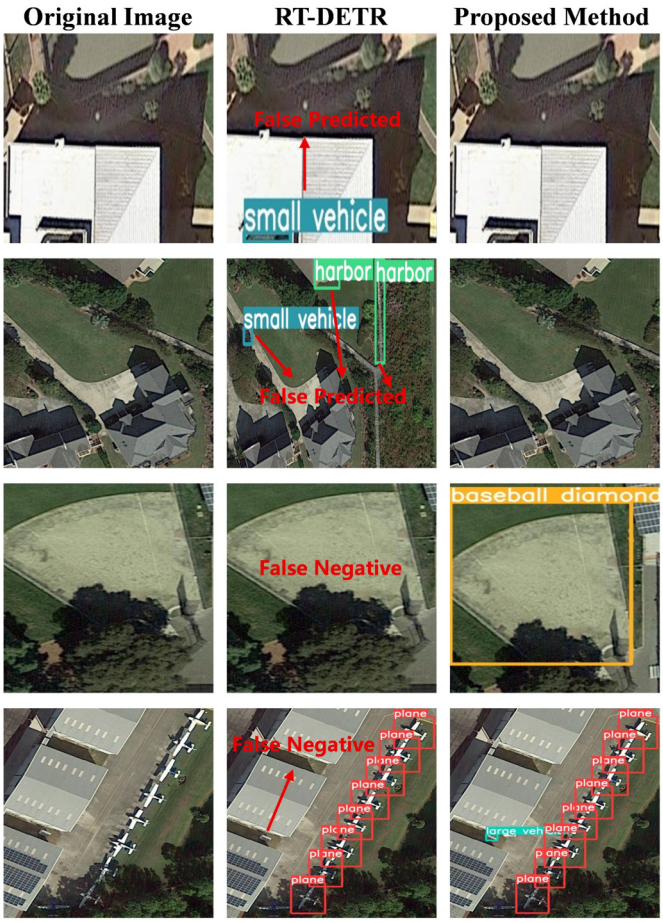


Fig. 10. Visualizations of detection results in DOTAv1.5.

H. Visualization

In order to observe the enhancement of the proposed method, the detection results are visualized on HRSC2016 and DOTAv1.5 in Figs. 9 and 10. The heatmaps are visualized with the GradCAMPlusPlus method [78].

Figs. 9 and 10 present the results of visualization without the original images. The second and third columns display the outcomes of the RT-DETR model and our proposed model,

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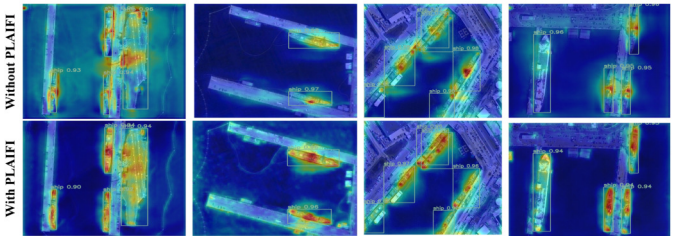


Fig. 11. Visualizations of PLAIFI on different feature maps. The first row is the visualization of the proposed method without PLAIFI, and the second row is the improvement of PLAIFI.

respectively. The fourth column illustrates the heatmap generated by our proposed model. The results depicted in Figs. 9 and 10 indicate that our method detects a greater number of correct targets while minimizing the detection of incorrect targets. This improvement demonstrates our proposed method's effectiveness in reducing boundary uncertainty and leveraging the central features of targets. Overall, our method exhibits a high level of attention to objects across both datasets, suggesting that the model possesses strong sensitivity in recognizing and focusing on object features, thereby enhancing its accuracy in identifying and classifying these objects.

Moreover, the proposed method achieves good results on both HRSC2016 (small-scale ships) and DOTAv1.5 (multiclass geospatial objects), which indicates that the proposed method provides robust solutions for detecting targets of varying scales in satellite imagery.

Furthermore, the improvements of PLAIFI on other feature maps are shown in Fig. 11. The first row is the visualization of our method without PLAIFI, and the second row is the improvement of PLAIFI. In different scenarios, our proposed approach allows the network to focus on useful information about the target, which demonstrates that the PLAIFI module is robust to feature map variance and significantly enhances feature expressiveness.

IV. Future Works

Building upon the current breakthroughs, three strategic directions are proposed for subsequent research development.

1) *Technical Architecture Evolution:* The framework's edge-aware paradigm could be extended through cross-modal attention mechanisms that integrate multisource remote sensing data (e.g., hyperspectral imagery and LiDAR point clouds) with RGB imagery via cross-branch masked attention operators, enhancing robustness under cloud occlusion through complementary feature alignment. An adaptive resolution selection module could dynamically adjust processing granularity based on target scales, optimizing computational resource allocation while preserving invariance across modalities. Furthermore, extending the PLAIFI attention mechanism to 3-D space would establish polarity-aware interaction models in elevation dimensions, enabling joint learning of geometric and spectral attributes from LiDAR and hyperspectral inputs. This evolution aligns with UrbanSAM's [79] core philosophy of multiscale invariance while addressing emerging challenges in multimodal urban analysis.
2) *Algorithmic Performance Optimization:* The inherent uncertainty in visual perception systems poses significant challenges for risk-sensitive decision scenarios, where prediction errors in object detection may lead to catastrophic consequences. Yelleni et al. [80] proposed Monte Carlo DropBlock, a Bayesian uncertainty modeling method, enhancing generalization, calibration, and uncertainty quantification in object detection under out-of-distribution scenarios. Building upon this uncertainty-aware paradigm, a Bayesian GEIE module could further quantify prediction uncertainties through probabilistic confidence estimation. This probabilistic reasoning framework proves particularly crucial for mission-critical systems like disaster early warning systems that demand rigorous risk assessment capabilities.
3) *Application Scenario Expansion:* The proposed framework is extensible to multihazard monitoring scenarios—including earthquake-induced landslide detection and postvolcanic-eruption debris flow tracking—where real-time, low-latency uncertainty-aware inference is critical. Future deployments will prioritize integration with edge computing devices and distributed sensor networks to address spatially heterogeneous hazards (e.g., urban flash floods and boreal forest wildfires) while maintaining computational efficiency.

V. Conclusion

This article presents a robust framework for object detection in remote sensing imagery, addressing critical challenges such as blurred boundary localization, extreme scale variations, and deployment computational inefficiency in Transformer-based detectors. By integrating the GEIE module and PLAIFI, the proposed method effectively propagates multiscale edge semantics and enhances intrascale feature representation. The GEIE module bridges the gap in cross-layer edge semantics through its PREFP module, recovering structural details lost during downsampling and significantly improving localization accuracy for targets of various sizes. Meanwhile, the polarity-aware decomposition of query–key interactions ensures comprehensive feature modeling while maintaining linear computational complexity, overcoming limitations of conventional attention mechanisms in cluttered aerial scenes.

Extensive experiments validate the framework's superiority, achieving state-of-the-art performance on HRSC2016, DOTAv1.5 datasets with 95.0% and 73.0% mAP, respectively, representing 4.0% and 5.9% absolute improvements over baseline RT-DETR. These results are attained with 63.5 GFLOPs computational complexity and 22M parameters, demonstrating a favorable accuracy-efficiency tradeoff where merely 11% GFLOPs/11% parameter growth yields over 25% relative accuracy gains. The framework exhibits remarkable adaptability to diverse targets, from small maritime vessels (HRSC2016) to large-scale geospatial objects (DOTAv1.5). The proposed methodology not only advances satellite image analysis through computationally sustainable

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innovation—achieving breakthrough accuracy within mainstream detector complexity ranges—but also provides a versatile foundation for time-sensitive applications in environmental sustainability monitoring, smart city infrastructure inspection, and climate change impact assessment, where balanced performance-efficiency requirements are critical.

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